Python

Readable steps over ceremony

THE MENTAL MODEL

A recipe that reads like a list of actions.

INPUT → LIST → LOOP → RESULT

READ THE CODE

```
prices = [10, 20, 30]
total = 0
for price in prices:
    total += price
print(total)  # 60
```

FOUR CLUES TO READ IT

- prices = [...] creates a list of values.
- for ... in ... visits each item in order.
- Indentation marks the loop body.
- print(...) displays the result.

WHAT MAKES IT DIFFERENT

- Types are checked as the program runs; optional type hints can help tools.
- Huge ecosystem for automation, data analysis and machine learning.

WHERE IT FITS

Automation, prototypes, data analysis, artificial intelligence (AI).

WATCH OUT

Simple syntax does not guarantee fast execution or correct results.

TRY ALONE • 60 SEC

Change the second price to 50. Predict the new total before running.

JavaScript

The behavior layer of the web

THE MENTAL MODEL

A live page reacts to an event without reloading everything.

EVENT → FUNCTION → CHANGE → SCREEN

READ THE CODE

```
const values = [10, 20, 30];
const sum = values.reduce(
  (a, b) => a + b, 0
);
console.log(sum); // 60
```

FOUR CLUES TO READ IT

- const declares a binding you cannot reassign.
- [...] is an ordered array.
- (a, b) => ... defines a short function.
- console.log writes to the developer console.

WHAT MAKES IT DIFFERENT

- Runs in browsers and also in server runtimes such as Node.js.
- Promises help coordinate operations that finish later (asynchronous work).

WHERE IT FITS

Interactive websites, browser tools, server applications.

WATCH OUT

const does not make an array immutable; types can change at runtime.

TRY ALONE • 60 SEC

Open a browser developer console; type 2 + 3 and observe 5.

TypeScript

JavaScript with type checking before running

THE MENTAL MODEL

A blueprint catches mismatched connectors before the device starts.

CODE → TYPE CHECK → JAVASCRIPT → RUN

READ THE CODE

```
const values: number[] = [10,20,30];
function add(xs: number[]): number {
  return xs.reduce((a,b) => a+b, 0);
}
console.log(add(values));
```

FOUR CLUES TO READ IT

: number[] requires an array of numbers.

function add(...) declares a named function.

: number describes the returned value.

Type annotations are erased when converted to JavaScript.

WHAT MAKES IT DIFFERENT

Checks many mistakes before code runs, but does not validate external data at runtime.

Uses the JavaScript runtime; TypeScript is not a faster execution engine.

WHERE IT FITS

Large web applications, shared codebases, editor assistance.

WATCH OUT

A type declaration is not a security check on input received from a network.

TRY ALONE • 60 SEC

Replace one number with "hello". What error appears before running?

C

Meet memory and the machine

THE MENTAL MODEL

You specify small operations and must watch every memory boundary.

SOURCE → COMPILE → MEMORY → PROGRAM

READ THE CODE

```
#include <stdio.h>
int main(void) {
  int x[3] = {10, 20, 30};
  int total = x[0] + x[1] + x[2];
  printf("%d\n", total);
  return 0;
}
```

FOUR CLUES TO READ IT

- #include adds declarations from a header.
- int is an integer type; x[0] is the first element.
- printf prints formatted text.
- main is the program entry point here.

WHAT MAKES IT DIFFERENT

- Explicit memory addresses and pointers enable precise low-level work.
- Compiled native programs are common in operating systems and embedded devices.

WHERE IT FITS

Firmware, operating-system components, hardware-adjacent code.

WATCH OUT

Out-of-bounds access can corrupt memory; the programmer must prevent it.

TRY ALONE • 60 SEC

Why is x[3] invalid for an array with three elements?

C++

Build abstractions without giving up control

THE MENTAL MODEL

Design reusable machines, but know who owns each component.

TYPES → OBJECTS → COMPILE → EXECUTE

READ THE CODE

```
#include <iostream>
#include <vector>
int main() {
  std::vector<int> xs{10,20,30};
  for (int x : xs) std::cout << x << " ";
}
```

FOUR CLUES TO READ IT

- #include brings library declarations.
- std::vector<int> is a resizable list of integers.
- for (int x : xs) traverses the list.
- std::cout sends text to standard output.

WHAT MAKES IT DIFFERENT

- Templates and classes enable reusable, strongly typed designs.
- Resource Acquisition Is Initialization (RAII) ties cleanup to object lifetime.

WHERE IT FITS

Game engines, graphics, performance-critical tools and libraries.

WATCH OUT

Powerful abstractions do not automatically prevent memory errors.

TRY ALONE • 60 SEC

Change std::vector<int> to std::vector<double>; what changes?

Java

Portable applications with classes and a managed runtime

THE MENTAL MODEL

Write a recipe in a standard format; a runtime follows it on many systems.

SOURCE → BYTECODE → RUNTIME → RESULT

READ THE CODE

```
class Main {
  public static void main(String[] args) {
    int[] xs = {10,20,30};
    int sum = 0;
    for (int x : xs) sum += x;
    System.out.println(sum);
  }
}
```

FOUR CLUES TO READ IT

- class groups code around a named type.
- main is the example entry method.
- for (int x : xs) visits each array element.
- System.out.println prints one line.

WHAT MAKES IT DIFFERENT

- Usually executes via the Java Virtual Machine (JVM), a managed runtime.
- Garbage collection reclaims many unused objects automatically.

WHERE IT FITS

Enterprise systems, backend services, Android legacy ecosystems.

WATCH OUT

Portable does not mean every platform setup or library is identical.

TRY ALONE • 60 SEC

Change the array to {3,4,5}; predict the printed sum.

C#

Expressive applications on the .NET platform

THE MENTAL MODEL

Objects, type checks and libraries assembled into practical services.

C# CODE → .NET RUNTIME → LIBRARIES → APP

READ THE CODE

```
int[] xs = {10, 20, 30};
int sum = 0;
foreach (int x in xs) {
    sum += x;
}
Console.WriteLine(sum);
```

FOUR CLUES TO READ IT

- int[] declares an integer array.
- foreach traverses the collection.
- sum += x adds x to the total.
- Console.WriteLine prints a line (console project).

WHAT MAKES IT DIFFERENT

- C# uses the .NET platform: managed runtime and broad libraries.
- Supports object-oriented, functional and asynchronous styles.

WHERE IT FITS

Business services, cross-platform apps, games with Unity.

WATCH OUT

Library or framework choices matter as much as the language syntax.

TRY ALONE • 60 SEC

Replace foreach with a for loop; what extra detail must you track?

Go

Small building blocks for concurrent services

THE MENTAL MODEL

Many independent workers can communicate through channels.

REQUEST → GOROUTINES → CHANNELS → SERVICE

READ THE CODE

```
package main
import "fmt"
func main() {
    values := []int{10, 20, 30}
    for _, x := range values {
        fmt.Println(x)
    }
}
```

FOUR CLUES TO READ IT

- package groups related Go files.
- := declares and initializes a local variable.
- range visits a collection; _ ignores the index.
- fmt.Println prints each value.

WHAT MAKES IT DIFFERENT

- Goroutines are lightweight concurrent functions; channels can coordinate them.
- Simple tooling and static binaries are convenient in service deployment.

WHERE IT FITS

Network servers, command-line tools, cloud infrastructure.

WATCH OUT

Concurrency needs coordination; goroutines do not remove data races.

TRY ALONE • 60 SEC

Which of the two values from range is ignored, and why?

Rust

Memory safety enforced through ownership

THE MENTAL MODEL

A library book has one owner; temporary readers follow clear rules.

VALUE → OWNER → BORROW → CHECK

READ THE CODE

```
fn main() {
    let values = [10, 20, 30];
    let sum: i32 = values.iter().sum();
    println!("{sum}");
}
```

FOUR CLUES TO READ IT

- fn declares a function.
- let binds a name to a value.
- iter().sum() traverses and adds values.
- println! is a macro for printing.

WHAT MAKES IT DIFFERENT

Ownership and borrowing restrict unsafe memory patterns at compile time.

No ordinary garbage collector is required for this memory model.

WHERE IT FITS

Systems components, secure services, high-performance tools.

WATCH OUT

The compiler can reject code that works in other languages until ownership is clarified.

TRY ALONE • 60 SEC

Why might two writers to the same mutable value be dangerous?

SQL

Ask for data, not the procedure to find it

THE MENTAL MODEL

Tell the librarian which books you want; they choose the search path.

QUESTION → QUERY → PLANNER → ROWS

READ THE CODE

```
SELECT department, COUNT(*) AS n
FROM employees
WHERE active = TRUE
GROUP BY department
ORDER BY n DESC;
```

FOUR CLUES TO READ IT

- SELECT names columns to return.
- FROM identifies the table.
- WHERE filters rows before grouping.
- GROUP BY / ORDER BY aggregate and sort results.

WHAT MAKES IT DIFFERENT

Structured Query Language (SQL) is declarative: describe the result.

Database engines choose execution plans and use indexes when useful.

WHERE IT FITS

Querying relational databases, reporting, data transformations.

WATCH OUT

The example assumes an employees table; order is not guaranteed without ORDER BY.

TRY ALONE • 60 SEC

Change DESC to ASC. Which departments appear first?

Bash

Connect programs in a command-line workflow

THE MENTAL MODEL

A conductor tells specialized tools what to do in sequence.

FILES → SHELL → COMMANDS → OUTPUT

READ THE CODE

```
for file in *.txt; do
  [ -f "$file" ] || continue
  echo "Found: $file"
done
```

FOUR CLUES TO READ IT

- for ... do ... done repeats commands for files.
- *.txt matches names ending in .txt.
- [-f "$file"] tests whether it is a file.
- echo prints; quotes preserve spaces in names.

WHAT MAKES IT DIFFERENT

Bash (Bourne Again Shell) orchestrates existing programs, not just calculations.

Pipes (|) connect standard output of one command to another.

WHERE IT FITS

Automation, developer workflows, server maintenance (with care).

WATCH OUT

Quoting and destructive commands matter: never run scripts you do not understand.

TRY ALONE • 60 SEC

Create two harmless .txt files; predict what the loop prints.

Kotlin

Safer concise code for applications

THE MENTAL MODEL

Describe what may be missing before it causes a failure.

VALUES → NULL CHECK → FUNCTION → APP

READ THE CODE

```
fun main() {
    val names = listOf("Ana", "Dan")
    for (name in names) {
        println(name)
    }
}
```

FOUR CLUES TO READ IT

fun declares a function.

val binds a read-only reference.

listOf constructs a list.

for (... in ...) traverses each entry.

WHAT MAKES IT DIFFERENT

Nullability is part of its type system; missing values are explicit.

Used with the Java Virtual Machine (JVM) and other targets.

WHERE IT FITS

Android applications, backend software, cross-platform projects.

WATCH OUT

val prevents reassignment, not all mutation inside the object.

TRY ALONE • 60 SEC

Why is val different from a promise that an entire list never changes?

Swift

Expressive programs with explicit optional values

THE MENTAL MODEL

A sealed box tells you whether there is something inside.

INPUT → OPTIONAL → CHECK → OUTPUT

READ THE CODE

```
let values = [10, 20, 30]
let sum = values.reduce(0, +)
print(sum)
let nickname: String? = nil
print(nickname ?? "anonymous")
```

FOUR CLUES TO READ IT

- let binds a value that cannot be reassigned.
- reduce(0, +) accumulates a sum from zero.
- String? is an optional string value.
- ?? supplies a fallback when the value is nil.

WHAT MAKES IT DIFFERENT

- Strong types, optionals and value semantics support safer app design.
- Common in the Apple ecosystem, including iOS and macOS.

WHERE IT FITS

Apple applications, mobile interfaces, native platform features.

WATCH OUT

An optional requires handling the absence of a value explicitly.

TRY ALONE • 60 SEC

Replace nil with "Vlad". Predict the printed text.

PHP

Server-side pages made from data

THE MENTAL MODEL

The server prepares an answer before sending it to your browser.

REQUEST → PHP CODE → SERVER → HTML

READ THE CODE

```
<?php
$prices = [10, 20, 30];
$sum = array_sum($prices);
echo $sum;  // 60
```

FOUR CLUES TO READ IT

- <?php starts a PHP code block.
- $prices is a variable; [] constructs an array.
- array_sum adds numeric array values.
- echo outputs text into the response.

WHAT MAKES IT DIFFERENT

PHP (PHP: Hypertext Preprocessor) runs on the server, not in browser JavaScript.

Can mix templates with backend logic; frameworks organize large applications.

WHERE IT FITS

Web backends, content-managed websites, server-generated pages.

WATCH OUT

Always escape untrusted output and validate input; echo alone is not safe HTML.

TRY ALONE • 60 SEC

What is sent to the browser after this script executes?

R

Think in columns, vectors and statistical models

THE MENTAL MODEL

Operate on a whole column instead of one number at a time.

DATA → VECTOR → SUMMARY → PLOT

READ THE CODE

```
scores <- c(10, 20, 30)
average <- mean(scores)
print(average)  # 20
print(scores > 15)
```

FOUR CLUES TO READ IT

<- assigns a value to a name.

c(...) combines values into a vector.

mean calculates an average.

scores > 15 compares all vector elements.

WHAT MAKES IT DIFFERENT

Vectorized operations make statistical exploration concise.

Rich ecosystem for analysis, visualization and research workflows.

WHERE IT FITS

Statistics, research, exploratory data analysis, reporting.

WATCH OUT

Missing values (Not Available, NA) and vector recycling can surprise beginners.

TRY ALONE • 60 SEC

Change a score to 90. Predict the mean and the comparisons.